

2f. Content of Discrete Mathematics (Optional paper Y544)

Introduction to Discrete Mathematics.

Discrete Mathematics is the part of mathematics dedicated to the study of discrete objects. Learners will study both pure mathematical structures and techniques and their application to solving real-world problems of existence, construction, enumeration and optimisation. Areas studied include counting, graphs and networks, algorithms, critical path analysis, linear programming and game theory.

7.06 Graphical Linear Programming

The concept of linear programming is explored as a tool for optimisation, including integer programming. Linear programmes are solved graphically.

7.07 The Simplex Algorithm

The simplex algorithm is used to solve optimisation problems.

7.08 Game Theory

Problems of conflict and cooperation are explored